Robust Multi-Modal Academic Integrity: An Integrated Vision-Language and Stylometric Framework for Handwritten and Digital Assignment Plagiarism Detection

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Abstract—*The proliferation of large language models (LLMs) such as GPT-4o, Claude 3.5, and Llama 3 has precipitated an unprecedented integrity crisis across educational institutions. Existing academic integrity frameworks suffer from three systemic deficiencies: (1) an inability to ingest non-digital modalities (e.g., handwritten assignments, scans, photos); (2) high false-positive rates on authentic human writing—particularly penalizing non-native English learners; and (3) severe data-privacy and FERPA/GDPR violations stemming from cloud-based text transmission. We propose an enterprise-grade, multi-modal, privacy-preserving academic integrity framework that couples an 8-layer layout-aware OCR vision pipeline with an adversarial defense-in-depth detection architecture. The vision engine utilizes CRAFT line detection, Microsoft TrOCR for cursive handwriting, and GOT-OCR 2.0 for mathematical notations and tabular formatting, reconciled through Levenshtein consensus arbitration and topological reading-order sorting. Extracted text is evaluated via a multi-layer attribution engine comprising: (a) a semantic curriculum alignment gate; (b) a local neural language model (Qwen 2.5 0.5B) computing token-level perplexity, loss variance, Shannon entropy, and sentence burstiness coefficient of variation; (c) an 18-dimensional stylometric, syntactic, and structural feature extractor; and (d) a calibrated ensemble Random Forest classifier. Concurrently, a dual-tier peer plagiarism subsystem detects image-level resubmissions via 64-bit difference perceptual hashing (dHash) and textual collusion via cohort-isolated unigram/bigram TF-IDF cosine similarity. Rigorous empirical evaluation across a benchmark dataset of 2,739 authentic student and LLM-generated submissions demonstrates that our model achieves **97.43% classification accuracy**, **99.22% precision**, **96.96% recall**, **98.08% F1-score**, and an **ROC-AUC of 0.9924**. Crucially, the system reduces the false-positive rate on authentic human writing to **1.59%** (3 false positives out of 189 test instances), while executing completely on-premise without cloud data exfiltration.*

Keywords—*Academic Integrity, AI-Generated Text Detection, Perplexity, Stylometry, Optical Character Recognition, TrOCR, GOT-OCR 2.0, Peer Plagiarism, Perceptual Hashing, Machine Learning.*

1. Introduction & Engineering Contributions

The rapid democratization of GenAI has created an urgent imperative for reliable, privacy-preserving academic integrity systems. Our platform introduces five foundational contributions:

- **End-to-End Multi-Modal Architecture:** Ingestion of digital documents (.pdf, .docx, .txt) and smartphone photos/scans of student cursive handwriting.
- **8-Layer Layout-Aware OCR Vision Pipeline:** CRAFT text detection, Microsoft TrOCR for handwriting, GOT-OCR 2.0 for math/tables, Levenshtein consensus arbitration, and topological column reconstruction.
- **Defense-in-Depth AI Attribution Engine:** Local Qwen 2.5 0.5B neural perplexity and sentence burstiness CV combined with an 18-dimensional stylometric feature vector and calibrated ensemble Random Forest classifier.
- **Dual-Tier Peer Collusion Engine:** Localized peer-to-peer plagiarism detection using 64-bit dHash perceptual hashing and cohort-isolated TF-IDF cosine similarity with MD5 matrix caching.
- **Empirical Rigor & Open Governance:** 97.43% test accuracy, 99.22% precision, 98.08% F1-score, and 0.9924 ROC-AUC, suppressing human false positives to 1.59% under complete on-premise execution.

2. System Architecture & Operational Quality Gates

Table 1 details the master operational threshold configuration across the vision and detection pipelines, enforcing strict quality gates before downstream tensor allocation.

Pipeline Stage	Operational Metric	Threshold / Limit	Action on Violation
Ingestion Quality	Laplacian Sharpness Var($\nabla^2 I$)	≥ 45.0	Reject blurry upload; prompt resubmission

Deskew Tolerance	Hough Line Angle $\|\theta\|$	$\leq 45.0°$	Affine rotation; reject if upside down
Recognition Router	TrOCR Beam Log-Prob	≥ 0.80	Escalate to GOT-OCR 2.0 / Qwen2-VL
Result Arbitration	Levenshtein Consensus	≥ 0.82	Consensus accepted; else domain weights
Human Review Flag	Minimum Confidence	≥ 0.70	Flag needs_review = True in UI
Curriculum Gate	Syllabus Cosine Similarity	≥ 0.12	Reject off-topic / fraudulent upload
Global Legibility	Overall Document OCR Score	$\geq 65.0\%$	Reject submission if < 50.0%

3. Empirical Benchmark Results (N=584 Held-Out Test Samples)

Metric	Training Partition (N=2,155)	Held-Out Test Set (N=584)
Classification Accuracy	100.00%	97.43%
Precision (AI Detection)	—	99.22%
Recall (AI Detection)	—	96.96%
F1-Score (AI Detection)	—	98.08%
ROC-AUC Score	—	0.9924
Specificity (Human True Negative Rate)	—	98.41%
False Positive Rate (FPR on Human Text)	—	1.59% (3 / 189 instances)

Ground Truth Class	Predicted Human (0)	Predicted AI (1)	Total Cohort
Authentic Human (0)	TN = 186 (98.41%)	FP = 3 (1.59%)	189 Essays
AI-Generated (1)	FN = 12 (3.04%)	TP = 383 (96.96%)	395 Essays

Top Discriminating Features (Gini Impurity Weight):

Rank	Feature Identifier	Weight	Primary Discriminative Domain
1	avg_paragraph_length	0.2632	Rigid Structural Symmetry (AI defaults to 4-5 sentences)
2	mean_perplexity	0.1668	Token Likelihood Surprise (LLM text exhibits low PPL 15-38)
3	entropy	0.1568	Shannon Information Density & Lexical Predictability
4	flesch_reading_ease	0.1099	Undergraduate Readability Clustering vs. Human Dispersion
5	perplexity_variance	0.0631	Monotonous Loss Fluctuation vs. Spiky Human Prose
6	stop_word_ratio	0.0571	Syntactic Scaffolding & Connector Uniformity
7	type_token_ratio	0.0392	Vocabulary Richness & Idiosyncratic Repetition
8	sentence_length_std	0.0274	Sentence Burstiness (Low std in AI, high std in human)

OCR Vision Benchmark & Hardware Resource Budget:

Architecture / Stack	CER (%) ↓	WER (%) ↓	Formula LaTeX Conversion	Two-Column Layout Acc.
Tesseract 5.0 (Baseline)	24.8%	38.2%	Fail (Unreadable)	41.2% (Interleaved columns)
EasyOCR (Baseline)	18.4%	29.1%	Fail (Unreadable)	58.6% (Misordered gutters)
Our Multi-Tier Pipeline	3.9%	6.2%	94.1% Syntactically Valid	98.4% Correct Topological Order

Compute Budget & Latency Profile (Single-Column Bounded):

Component	Params	Memory	Device	Latency
OpenCV Preprocessor	—	~50 MB RAM	CPU	180 ms / page
CRAFT Text Detector	20.7 M	~180 MB VRAM	CUDA	450 ms / page
TrOCR Base	334.0 M	~1.3 GB VRAM	CUDA	110 ms / line
GOT-OCR 2.0	580.0 M	~2.2 GB VRAM	CUDA	1.6 s / region
Qwen 2.5 LM	490.0 M	~1.1 GB VRAM	CUDA	320 ms / essay
Random Forest Classifier	200 Trees	~15 MB RAM	CPU	< 4 ms / vector
TF-IDF Plagiarism	5k Vocab	~30 MB RAM	CPU	22 ms / cohort

4. Institutional Governance & Conclusion

By combining an 8-layer layout-aware OCR vision pipeline with an adversarial defense-in-depth detection architecture (Qwen 2.5 0.5B perplexity, 18-dimensional stylometry, and calibrated Random Forest) and localized peer plagiarism detection, the platform provides an accurate, equitable, and privacy-preserving foundation for academic institutions. All processing executes on-premise without cloud data exfiltration, ensuring strict compliance with FERPA and GDPR regulations.